Bios 7xx: Deep Learning Methods for Biomedical Applications with Pytorch

Syllabus for Bios 7XX, Spring 2025. 3 Credits

Course Description and Goals:

In the rapidly evolving fields of computational biology and biomedical research, the advent of deep learning methodologies has revolutionized how data-driven discoveries are made. This course, "Deep Learning Methods in Biomedical Sciences with PyTorch", is an in-depth exploration into the synergistic interplay of advanced computational methods and the intricate matrices of biomedical data.

As datasets grow in size and complexity, traditional algorithms often fall short in capturing the nuanced relationships and patterns within the data. Deep learning, with its multi-layered neural networks and capacity for learning from vast amounts of data, offers promising solutions for some of the most pressing challenges in the biomedical field.

Spanning from fundamental neural network architectures to cutting-edge models tailored for biomedical applications, this course is structured to provide students with both the theoretical knowledge and practical skills necessary to spearhead innovative research endeavors. By using PyTorch, a leading deep learning framework known for its flexibility and efficiency, students will be equipped with the hands-on experience of implementing, training, and evaluating deep learning models.

The curriculum is meticulously designed to start with foundational algorithms, ensuring students grasp the underpinnings of neural computations. From there, it scales progressively into advanced topics such as convolutional neural networks for medical image analysis, recurrent networks for sequential health data, transformers for genomic sequences, and specialized models like BioBERT for clinical notes.

But this course is more than just algorithms and coding. It places a strong emphasis on real-world applications, ensuring students understand the broader implications of their work. Through case studies, project work, and interactive discussions, students will delve into challenges like disease prediction, medical image segmentation, and genomic data interpretation, appreciating the potential of deep learning in shaping the future of biomedical research.

"Deep Learning Methods in Biomedical Sciences with PyTorch" is not just a course—it's an invitation to be at the forefront of biomedical innovation, harnessing the power of deep learning to decipher the complexities of life and health.

Instructor: Dr. Hongtu Zhu is a tenured professor of biostatistics, statistics and operation research, genetics and computer science at University of North Carolina at Chapel Hill (UNC-CH). He was DiDi Fellow and Chief Scientist of Statistics at DiDi Chuxing, an IOT company and was Endowed Bao-Shan Jing Professorship in Diagnostic Imaging at MD Anderson Cancer Center. He is an internationally recognized expert in statistical learning, medical image analysis, precision medicine, biostatistics, artificial intelligence, and big data analytics.

Class Time:

Class Location:

Office number: 919-966-7272.

Office Hours: Friday 12:00pm-1:00pm.

My websites: https://bigkp.org/ and https://www.med.unc.edu/bigs2/

Course website:

Appointment.

Please make appointments.

If you have any small issues, such as typos in the note and homework assignment, please send me/TA an email. I would love to chat with you through email.

e-mail: htzhu@email.unc.edu

Course Format: The course format will consist of a seminar-style class that meets weekly. Students are expected to complete the readings before class and come to each class prepared to discuss the texts (with the exception of week 1, where the readings will be done after the fact). Lectures will be supplemented with additional readings and examples from the literature.

Course-at-a-Glance: The instructor reserves the right to make changes to the syllabus, including topics, readings, assignments, and due dates. Any changes will be announced as early as possible.

Prerequisites: Basic Python programming. Concepts in linear algebra, probability, and calculus are advantageous. Both master students and ph.d students are welcame to register.

Teaching Asistant:

Graded Assignments: This course will include the following graded assignments that contribute to your final grade in the course. Such graded assignments include seven homeworks and a report and presentation for a medical imaging related dream challenge.

Grading System: Final grade is based on the performance of seven homeworks (7×10=70 points) and the dream challenge report and representations (30 points). **I will give a personal award to a student who will rank the top in a selected related dream challenge.** The grades reported will be transformed into HPF scale (H: 90-100; P: 70-90; LP: 60-69; F: 0-59).

Use of Generative AI: Unless otherwise specified, AI Use is generally allowed for many tasks. It requires that students document how they used AI.

No middle term and final exams.

Course Modules

1. Introduction to Deep Learning, PyTorch & Basic Algorithms

- What is machine learning and deep learning? Jiarui
- Basics of Supervised Learning: Linear Regression and Logistic Regression
- Introduction to PyTorch
- PyTorch Basics: Tensors, Autograd, Neural Network module

2. Neural Networks Fundamentals

- Perceptron Model and Multilayer Perceptrons (MLP) Jiarui
- Activation Functions
- Loss Functions and Optimization Techniques

3. Convolutional Neural Networks (CNN)

Xiaoqi

- Understanding the Convolution Operation
- Pooling, Stride, and Padding
- Biomedical Application: Medical Image Classification

ADNI or UKB — clinical outcome
use 1k subjects hippocampus mtx (Yalin) - CNN
AD proxy and fluid intelligence
start with 2D -> 3D CNN

4. Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM)

Shuai

- Basics of Sequence Models and their Limitations
- Introduction to LSTMs and GRUs
- Biomedical Application: Analyzing Sequential Health Data

predict Parkinson / Cardiovascular -
Arctherometer data from UKB

5. Graph Convolutional Networks (GCNs)

Shuai

- Basics of graphs and graph neural networks
- Designing a GCN with PyTorch
- Biomedical applications: Protein-protein interaction prediction

Pytorch Geometric package
could change to structural / functional connectome
HCP data - ADNI UKB - clinical scope

6. Generative Adversarial Networks (GAN)

Xiaoqi

- Understanding Generative Models
- Basics of GAN and its Architectures
- Variational Autoencoder
- Biomedical Application: Generating Synthetic Medical Images

ADNI PET and MRI (DTI) image imputation (1,2 as training, 3 as validation)
apply to UKB - impute PET data (not available in UKB)

7. Transformers and Attention Mechanisms

Runpeng, Minghu

- Basics of the Transformer Architecture and Attention Mechanism
- Self-Attention and Positional Encoding
- Biomedical Application: DNA Sequence Analysis

minghu: multimodal & segmentation

8. Text Mining in Biomedical Sciences

Yue

- Basics of Text Mining and Natural Language Processing
- Word Embeddings and Tokenization
- Biomedical Application: Extracting Information from Medical Research Papers

9. BioBERT and ClinicalBERT

Yue

- Introduction to BERT and its Variants
- Application in Biomedical Text Mining text transformer - attention model
- Biomedical Application: Clinical Note Analysis

10. Image Segmentation and Registration

Minghu

- Techniques for Image Segmentation
- Introduction to Image Registration segmentation + DL registration brain/heart example from UKB
- Biomedical Application: Tumor Localization in Radiology Images

11. Spatio-temporal Process

Shuai

- Basics of Spatio-temporal Data G360? UKB clinical trial EHR ICD code - longitudinal by nature
- Modeling Techniques for Spatio-temporal Prediction
- Biomedical Application: Predicting the Spread of Diseases over Time and Space

12. Diffusion Methods

Runpeng

- Introduction to Diffusion Processes
- Deep Learning Approaches to Diffusion
- Biomedical Application: Cell Type Classification based on Diffusion Patterns

Copyright of Python and Pytorch codes and datasets

All Python and Pytorch codes and datasets presented in this lecture note are copied from various books and websites. Please use them for personal purpose.

Reference Books

- *Deep Learning* by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
- *Python Deep Learning* by Ivan Vasilev and Daniel Slater.
- *Deep Learning for Medical Image Analysis* by S. Kevin Zhou, Hayit Greenspan, and Dinggang Shen.
- *Bioinformatics Data Skills* by Vince Buffalo.
- *Pattern Recognition and Machine Learning* by Christopher Bishop.
- *Neural Network Methods in Natural Language Processing* by Yoav Goldberg.
- *Practical NLP: A Comprehensive Guide to Building Real-World NLP Systems* by Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, and Harshit Surana.

Tips for Success:

(i) Read lecture slices and assigned reading materials carefully before and after class.

(ii) It is critical to solve all homework problems independently.

(iii) Ask TA and instructor directly for helps when you feel uncomfortable with some course materials.

Attendance/ Participation Your attendance and active participation are an integral part of your learning experience in this course. If you are unavoidably absent, please notify the course instructor (and Teaching Assistant if one is assigned). No right or privilege exists that permits a student to be absent from any class meetings, except for these University Approved Absences, including Authorized University activities, Disability/religious observance/pregnancy, as required by law and approved by Accessibility Resources and Service (https://ars.unc.edu) and/or the Equal Opportunity and Compliance Office (https://eoc.unc.edu/what-we-do/accommodations/), and Significant health condition and/or personal/family emergency as approved by the Office of the Dean of Students (https://odos.unc.edu/), Gender Violence Service Coordinators (https://womenscenter.unc.edu/resources/gender-violence-services/), and/or the Equal Opportunity and Compliance Office (https://eoc.unc.edu/what-we-do/accommodations/).

Honor Code: As a student at UNC Chapel Hill, you are bound by the university's Honor Code, (https://catalog.unc.edu/policies-procedures/honor-code/), through which UNC maintains standards of academic excellence and community values. It is your responsibility to learn about and abide by the code. To ensure an effective Honor System at UNC, in this course students are expected to:

- · Conduct all academic work within the letter and spirit of the Honor Code, which prohibits the giving or receiving of unauthorized aid in all academic processes.
- · Learn and apply the recognized techniques of proper attribution of sources used in written work; and to identify allowable resource materials or aids to be used during completion of any graded work.
- · Students may use materials they wrote for other courses, but only if the student themselves produced the work.
- · For homework, students may verbally discuss approaches to the problems but each student should independently write up the answer and verify solutions.
- · For take-home tests, students must work completely independently without communicating with other students, tutors, or anyone else about any material related to the test questions. The test is 'open book' and 'open notes.'

If you have any questions about your rights and responsibilities, consult the Office of Student Conduct (https://studentconduct.unc.edu/) or review the following resources: Honor System (https://studentconduct.unc.edu/honor-system; Honor System module (https://studentconduct.unc.edu/honor-system-module); UNC Library's plagiarism tutorial (https://guides.lib.unc.edu/plagiarism); UNC Writing Center's handout on plagiarism (https://writingcenter.unc.edu/tips-and-tools/plagiarism/).

Inclusive Excellence:

In this class, we practice the Gillings School's commitment to inclusion, diversity, anti-racism and equity in the following ways.

- Develop classroom participation approaches that acknowledge the diversity of ways of contributing in the classroom and foster participation and engagement of all students.

- Structure assessment approaches that acknowledge different methods for acquiring knowledge and demonstrating proficiency.
- Encourage and solicit feedback from students to continually improve inclusive practices.
- Treat all members of the Gillings community (students, faculty, and staff) as human persons of equal worth who deserve dignity and respect, even in moments of conflict and disagreement.
- Contribute to creating a welcoming and inclusive classroom environment, where all are able to learn and grow from one another.
- Acknowledge and respect the diversity of experiences that others bring to the classroom and the ways in which this richness enhances everyone's learning
- Strive to maintain a spirit of curiosity and generosity, particularly in the face of new and/or seemingly contradictory information and perspectives Encourage and solicit feedback from students to continually improve inclusive practices.

Student Feedback and Equity Concerns: The Gillings School has in place a mechanism for students to provide feedback, including specifically equity concerns and bias-related issues. You can use this form to describe feedback, both positive and negative, about anything including issues related to your experience as a student at Gillings, administrative processes, and classroom activities. This form will also allow you to specifically describe incidents in which racial or other equity-related bias, or microaggressions, occurred. You may submit this form anonymously. However, for us to follow up and provide the necessary support, we encourage you to include your contact information. For further information, please visit the Student Feedback and Equity Concerns FAQ. Please note that this form does not take the place of any University process or policy. If you would like to report an incident under the University's policy on Prohibited Discrimination, Harassment, and Related Misconduct including Sexual and Gender Based Harassment, Sexual Violence, Interpersonal Violence, and Stalking, please visit Safe At UNC or the Equal Opportunity and Compliance Office (EOC) for additional information, including resources, contact, and reporting options.

Technical Support: The best way to help prevent technical issues from causing problems for assignments and quizzes is to submit them at

least 24-36 hours before the due date and time. Your instructor cannot resolve technical issues, but it's important to notify them if you are experiencing issues. If you have problems submitting an assignment or taking a quiz in Sakai, immediately do the following:

- 1. Contact the UNC Information Technology Services (ITS) department with the time you attempted to do your course action and what the course action was.
- 2. Email your instructor with the information you sent to ITS and what time you sent the information.

The ITS department provides technical support 24-hours per day, seven days per week. If you need computer help, please contact the ITS Help Desk by phone at +1-919-962-HELP (4357), or by online help request at http://help.unc.edu/help/olhr, or by UNC Live Chat at http://help.unc.edu/chat.

Title IX at UNC Chapel Hill: Any student who is impacted by discrimination, harassment, interpersonal (relationship) violence, sexual violence, sexual exploitations, or stalking is encouraged to seek resources on campus or in the community. Please contact the Director of Title IX Compliance / Title IX Coordinator (Adrienne Allison, adrienne.allison@unc.edu), Report and Response Coordinators in the Equal Opportunity and Compliance Office (reportandresponse@unc.edu), Counseling and Psychological Services (confidential), or the Gender Violence Services Coordinators (gvsc@unc.edu; confidential) to discuss your specific needs. Additional resources are available at https://safe.unc.edu.